

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.1 Recursive and Iterative Computation Analyzer (10 marks)

```
def rec_fib(n):
    if n<=1:
        return n
    return fib(n-1)+fib(n-2)

def rec_fact(n):
    if n<=1:
        return 1
    return n*fact(n-1)

def iter_fib(n):
    a, b=0,1
    if n==0:
        return 0
    if n==1:
        return b
    else:
        while n>1:
            a, b = b, a+b
            n-=1
        return a

def iter_fac(n):
    a=1
    if a<=0:
        return 1
    else:
        while n>1:
            a=a*(a+1)
            n-=1
        return a

def analyze_recursive_iterative(n):
    recfib=rec_fib(n)
    recfact=rec_fact(n)
    iterfib=iter_fib(n)
    iterfac=iter_fac(n)
    print("Report")
    print("recursive fibonacci",recfib)
    print("iterative fibonacci",iterfib)
    print("recursive factorial",recfact)
    print("iterative factorial",iterfac)

    return []
```

Q2. 1.2 Linear Search and Binary Search Comparison (5 marks)

```
def linear_search(array, target):
    comparison=0
    for i in range(len(array)):
        comparison+=1
        if array[i]==target:
            return i, comparison

def binary_search(array, target):
    low=0
    comparison=0
    high=len(array)
    while low<high:
        comparison+=1
        mid=(low+high)//2
        elem=array[mid]
        if elem<target:
            low=mid
        elif elem>target:
            high=mid
        else:
            return mid, comparison
```

```
if elem==target:
    return mid, comparision
else:
    high=mid
```

```
def compare_search_algorithms(arr, target):
```

```
    return []
```